MECH 325
Homework Assignment #5
Due Dec 8

Problem 1 (Question 8-17)

Two identical aluminum plates are each 2 in thick, and are compressed with one bolt and nut. Washers are used under the head of the bolt and under the nut.
Washer properties: steel; ID = 0.531 in; OD = 1.062 in; thickness = 0.095 in Nut properties: steel; height = $\frac{7}{16}$ in Bolt properties: $\frac{1}{2}$ in-13 UNC grade 8

Plate properties: aluminum; E = 10.3 Mpsi; S_u = 47 kpsi; S_y = 25 kpsi

(*a*) Determine a suitable length for the bolt, rounded up to the nearest $\frac{1}{4}$ in.

(*b*) Determine the bolt stiffness.

(*c*) Determine the stiffness of the members

8-17 **a**) Grip, $l = 2(2 + 0.095) = 4.19$ in. $L \geq 4.19 + 7/16 = 4.628$ in.
Rounding up, $L = 4.75$ in *Ans.*

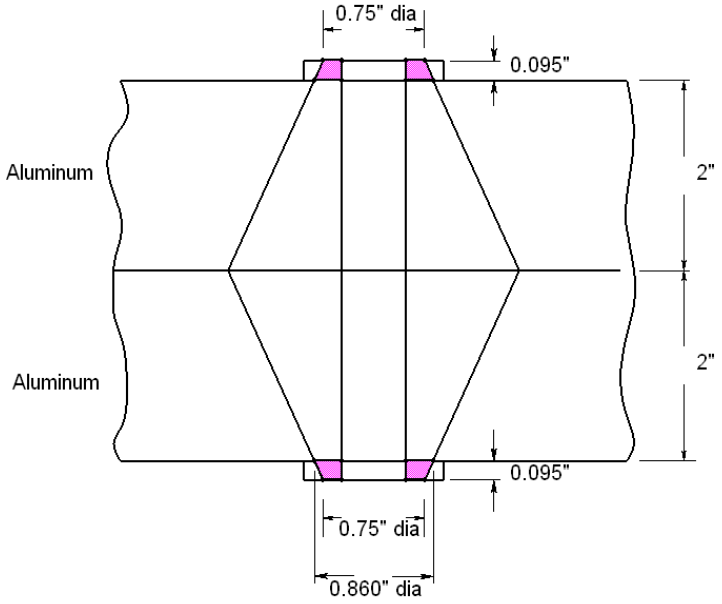
(**b**) From Eq. (8-13), $L_T = 2d + 1/4 = 2(0.5) + 0.25 = 1.25$ in

From Table 8-7, $l_d = L - L_T = 4.75 - 1.25 = 3.5$ in, $l_t = l - l_d = 4.19 - 3.5 = 0.69$ in

$A_d = \pi(0.5^2)/4 = 0.1963$ in^2. From Table 8-2, $A_t = 0.1419$ in^2. From Eq. (8-17)

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d} = \frac{0.1963(0.1419)30}{0.1963(0.69) + 0.1419(3.5)} = 1.322 \text{ Mlbf/in} \qquad \textit{Ans.}$$

(**c**)



Upper and lower halves are the same. For the upper half,
Steel frustum: t = 0.095 in, d = 0.531 in, D = 0.75 in, and E = 30 Mpsi. From Eq. (8-20)

$$k_1 = \frac{0.5774\pi(30)0.531}{\ln\dfrac{\left[1.155(0.095)+0.75-0.531\right](0.75+0.531)}{\left[1.155(0.095)+0.75+0.531\right](0.75-0.531)}} = 89.20 \text{ Mlbf/in}$$

Aluminum: t = 2 in, d = 0.5 in, D =0.75 + 2(0.095) tan 30° = 0.860 in, and E = 10.3 Mpsi. Eq. (8-20) $\Rightarrow$ k_2 = 9.24 Mlbf/in
For the top half, $k'_m = (1/k_1 + 1/k_2)^{-1}$ = $(1/89.20 + 1/9.24)^{-1}$ = 8.373 Mlbf/in
Since the bottom half is the same, the overall stiffness is given by

$$k_m = (1/k'_m + 1/k'_m)^{-1} = k'_m/2 = 8.373/2 = 4.19 \text{ Mlbf/in} \qquad \textit{Ans}$$

Problem 2 (Question 8-29 modified)

For a bolted assembly with six bolts, the stiffness of each bolt is k_b = 3 Mlbf/in and the stiffness of the members is k_m = 12 Mlbf/in per bolt. An external load of 80 kips is applied to the entire joint. Assume the load is equally distributed to all the bolts. It has been determined to use $\frac{1}{2}$ in-13 UNC grade 8 bolts with rolled threads. Assume the bolts are preloaded to 75 percent of the proof load.

(*a*) Determine the required preload (Assume assembly is non-permanent) and the torque required to reach the preload (Assume the bolts are lubricated). What size of wrench is required to tighten the nut?
(*b*) Determine the overload factor of safety.
(*c*) Determine the factor of safety based on joint separation.
(*d*) Assuming that the external load is repeated load (from 0 to P), determine the fatigue factor of safety using the Goodman criterion.
(e) Comment on the results.

$C = k_b / (k_b + k_m) = 3/(3+12) = 0.2$, $P = P_{\text{total}}/N = 80/6 = 13.33$ kips/bolt

Table 8-2, A_t = 0.141 9 in^2; Table 8-9, S_p = 120 kpsi; Eqs. (8-31) and (8-32),
$F_i = 0.75 A_t S_p = 0.75(0.141\,9)(120) = 12.77$ kips

(a) The nut size for the ½" 13-UNC bolt = 1.5(1/2")= ¾" (nominal wrench size is 1.5 bolt dia.)
Equation (8-27) estimates the preload $T = KF_i d$ where K is the bolt condition factor
From Table 8-15, K = 0.18 for lubricated nuts.

Hence T = 0.18 (12.77 kips) (0.5") = 1149 lb-in = 95.8 ft-lbs of torque *Ans.*

Note: This value is VERY high. A standard ¾" wrench is less than 10" in length. Say you apply the force at about 8" from the nut, you will need to apply approximately 140 pounds of arm strength on the wrench to achieve this preload.

(**b**) From Eq. (8-29), the overload factor is

$$n_L = \frac{S_p A_t - F_i}{CP} = \frac{120(0.141\,9) - 12.77}{0.2(13.33)} = 1.60 \qquad \textit{Ans.}$$

(**c**) From Eq. (8-30), the joint separation factor of safety is

$$n_0 = \frac{F_i}{P(1-C)} = \frac{12.77}{13.33(1-0.2)} = 1.20 \qquad \textit{Ans.}$$

d) Goodman Eq. (8-45) for grade 8 bolts, S_e = 23.2 kpsi (Table 8-17), S_{ut} = 150 kpsi (Table 8-9)

$$\sigma_i = \frac{F_i}{A_t} = \frac{12.77}{0.141\,9} = 90.0 \text{ kpsi}$$

Eq. (8-39), $$\sigma_a = \frac{CP}{2A_t} = \frac{0.2(13.33)}{2(0.141\,9)} = 9.39 \text{ kpsi}$$

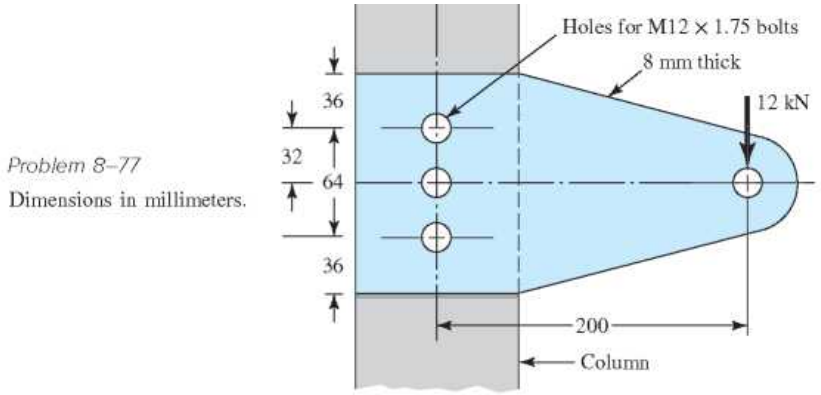
Eq. (8-41), $$\sigma_m = \sigma_a + \sigma_i = 9.39 + 90.0 = 99.39 \text{ kpsi}$$

$$n_f = \frac{S_e(S_{ut} - \sigma_i)}{\sigma_a(S_{ut} + S_e)} = \frac{23.2(150 - 90.0)}{9.39(150 + 23.2)} = 0.856 \qquad \textit{Ans.}$$

e) The factor of safeties are all low. The design will fail in fatigue. Since Grade 8 bolts are already prescribed, it is suggested that more bolts be used (say 8 bolts) or more to larger size bolt (say 5/8" diameter).

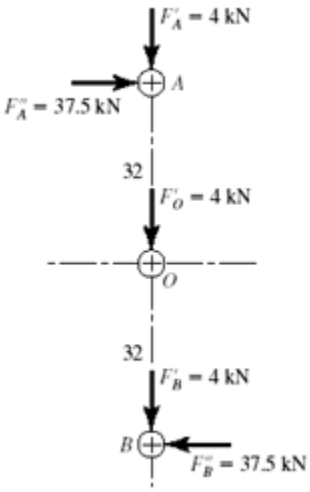
Problem 3 (Question 8-77)

The cantilever bracket is bolted to a column with three $M12 \times 1.75$ ISO 5.8 bolts. The bracket is made from AISI 1020 hot-rolled steel. Assume the bolt threads do not extend into the joint. Find the factors of safety for the following failure modes: shear of bolts, bearing of bolts, bearing of bracket, and bending of bracket (top of the bracket at the bolt centerline). Comment on the results



Problem 8–77
Dimensions in millimeters.

8-77 Bolts, from Table 8-11, S_y = 420 MPa
Bracket, from Table A-20, S_y = 210 MPa



$$F' = \frac{12}{3} = 4 \text{ kN};\ M = 12(200) = 2400 \text{ N} \cdot \text{m}$$
$$F''_A = F''_B = \frac{2400}{64} = 37.5 \text{ kN}$$
$$F_A = F_B = \sqrt{(4)^2 + (37.5)^2} = 37.7 \text{ kN}$$
$$F_O = 4 \text{ kN}$$

Bolt shear:
The shoulder bolt shear area, $A_s = \pi(12^2) / 4 = 113.1 \text{ mm}^2$

$$S_{sy} = 0.577(420) = 242.3 \text{ KPa}$$

$$\tau = \frac{37.7(10)^3}{113} = 333 \text{ MPa}$$
$$n = \frac{S_{sy}}{\tau} = \frac{242.3}{333} = 0.728 \qquad \textit{Ans.}$$

Bearing on bolts:

$$A_b = 12(8) = 96 \text{ mm}^2$$

$$\sigma_b = -\frac{37.7(10)^3}{96} = -393 \text{ MPa}$$

$$n = \frac{S_{yc}}{|\sigma_b|} = \frac{420}{393} = 1.07 \qquad \textit{Ans.}$$

Bearing on member:

$$\sigma_b = -393 \text{ MPa}$$

$$n = \frac{S_{yc}}{|\sigma_b|} = \frac{210}{393} = 0.534 \qquad \textit{Ans.}$$

Bending stress in plate:

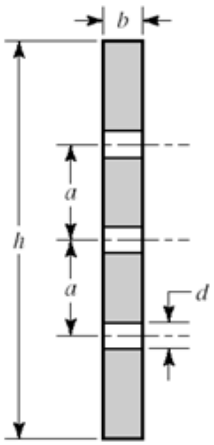
$$I = \frac{bh^3}{12} - \frac{bd^3}{12} - 2\left(\frac{bd^3}{12} + a^2bd\right)$$

$$= \frac{8(136)^3}{12} - \frac{8(12)^3}{12} - 2\left[\frac{8(12)^3}{12} + (32)^2(8)(12)\right]$$

$$= 1.48(10)^6 \text{ mm}^4 \qquad \textit{Ans.}$$

$$\sigma = \frac{Mc}{I} = \frac{2400(68)}{1.48(10)^6}(10)^3 = 110 \text{ MPa}$$

$$n = \frac{S_y}{\sigma} = \frac{210}{110} = 1.91 \qquad \textit{Ans.}$$



Failure is predicted for bolt shear and bearing on member.

Problem 4 (Mott Chapter 17)

An ACME thread power screw is loaded in tension with a force of 27000 lbs.

a) What size of screw should be used to maintain a tensile stress below 10,000 psi. (Table 17-1)?
b) What length of nut is required to maintain a shear stress in the threads less than 5000 psi.?
c) Using a thread friction coefficient of 0.15 and a collar friction of 0.06, determine the raising and lowering torque. Assume the collar friction radius is 2.0 inches.
d) What is the efficiency of the power screw?
e) If the input power is 10HP, determine the maximum speed you can raise the load.

Note: Similar solution from 2023

a) $\sigma = \frac{F}{A_t} \rightarrow A_t = \frac{F}{\sigma} = \frac{30000\ lbs}{10\,000\ lbs/in^2} = 3\ in^2$ - Select screw with $A_t > 3\ in^2$

[Table 17-1]

TABLE 17-1 Preferred Acme Screw Threads

Nominal major diameter, D (in)	Threads per in, n	Pitch, $p = 1/n$ (in)	Minimum minor diameter, D_r (in)	Minimum pitch diameter, D_t (in)	Tensile stress area, A_t (in^2)	Shear stress area, A_s (in^2)*
1/4	16	0.0625	0.1618	0.2043	0.026 32	0.3355
5/16	14	0.0714	0.2140	0.2614	0.044 38	0.4344
3/8	12	0.0833	0.2632	0.3161	0.065 89	0.5276
7/16	12	0.0833	0.3253	0.3783	0.097 20	0.6396
1/2	10	0.1000	0.3594	0.4306	0.1225	0.7278
5/8	8	0.1250	0.4570	0.5408	0.1955	0.9180
3/4	6	0.1667	0.5371	0.6424	0.2732	1.084
7/8	6	0.1667	0.6615	0.7663	0.4003	1.313
1	5	0.2000	0.7609	0.8726	0.5175	1.493
1 1/4	5	0.2000	0.8753	0.9967	0.6881	1.722
1 1/2	5	0.2000	0.9998	1.1210	0.8831	1.952
1 3/4	4	0.2500	1.0719	1.2188	1.030	2.110
2	4	0.2500	1.1965	1.3429	1.266	2.341
2 1/4	4	0.2500	1.4456	1.5916	1.811	2.803
2 1/2	4	0.2500	1.6948	1.8402	2.454	3.262
2 3/4	3	0.3333	1.8572	2.0460	2.982	3.610
3	3	0.3333	2.1065	2.2939	3.802	4.075
3 1/4	3	0.3333	2.3558	2.5427	4.711	4.538
3 1/2	2	0.5000	2.4326	2.7044	5.181	4.757
3 3/4	2	0.5000	2.5314	3.2026	7.388	5.700
4	2	0.5000	3.4302	3.7008	9.985	6.640
4 1/2	2	0.5000	3.9291	4.1991	12.972	7.527
5	2	0.5000	4.4281	4.6973	16.351	8.511

*Per inch of length of engagement.

Select:

2.5 in diameter

3 threads/inch

$p = 1/3$ in

$A_t = 3.802\ in^2$

$A_s = 4.075\ in^2$ (per 1 in engagement length)

b) $\tau = \frac{F}{A_s}$

$A_s > \frac{30000\ lb}{6000\ lb/in^2} > \frac{5\ in^2}{L} = \frac{4.075\ in^2}{1\ in}$

$L = \frac{5\ in^2}{4.075\ in^2} = 1.227\ in \rightarrow$ round to $1.25\ in$

Nut Length: **$L = 1.25$ in**

c) $f_t = 0.15$

$f_c = 0.06$ @ 2.0 in radius

RAISING TORQUE: COLLAR: $T_c = f_c \cdot F \cdot r_c = 0.06\,(30000\,lb)(2\,in)$

$T_c = 3600$ lb-in

THREAD: $T_R = \frac{F d_m}{2}\left(\frac{\pi f d_m \sec\alpha + l}{\pi d_m - f l \sec\alpha}\right)$

[Eq. 8-5]

l = Pitch = $1/3$ in

$d_m = \frac{d + d_r}{2} = \frac{2.5 + 2.1065}{2} = 2.30325\ in \rightarrow T_R = 6994.3$ lb-in

$\alpha = 29°/2 = 14.5°$

TOTAL RAISING TORQUE = 10594 lb-in

LOWERING TORQUE : Collar friction torque is same as raising : 3600 lb-in

$$T_L = \frac{F d_m}{2}\left(\frac{\pi f d_m \sec\alpha - l}{\pi d_m + f l \sec\alpha}\right) = 3734.6 \text{ lb-in}$$

TOTAL LOWERING TORQUE = 7335 lb-in

d) EFFICIENCY

[Eq. 8-4] $e = \frac{T_o}{T_R} = \frac{Fl}{2\pi T_R} = \frac{30\,000 \text{ lb}\left(\frac{1}{3} \text{ in}\right)}{2\pi\left(10594 \text{ lb-in}\right)} = 0.1502$

$e = 0.15 = 15\%$

e) MAX RAISING SPEED with 10 hp input power

$$T = \frac{63025 \cdot P}{n}$$

$$n = \frac{63025\,(10 \text{ hp})}{10594 \text{ lb-in}} = 59.49 \text{ rpm}$$

Convert to linear speed using lead:

$L = \frac{1}{3}$ in / revolution

$$V = 59.49 \frac{\text{rev}}{\text{min}} \times 0.333 \frac{\text{in}}{\text{rev}} = 19.83 \text{ in/min} \quad /12$$

$= 1.653$ fpm

$= 0.330$ in/s

$V = 0.33$ in/s $= 1.65$ feet/min.

Problem 5 (Mott Chapter 17)

A ball screw for a machine table drive is to be selected. The axial force to be transmitted by the screw is 900 lbs. The table moves 36 inches per cycle and is expected to cycle 12 times per hour, 365 days/year for 10 years.

a) Select the appropriate screw for the application (use Mott text).

b) Compute the torque required to drive the screw.

c) For a travel speed of 2 inches/sec compute the power required for the screw.

d) Use the Thomson Linear Motion guides to confirm your selection in part a)

e) For safety factor of 2.5, what is the maximum length of power screw you can select (Assume fixed ends on each side)? What is the maximum speed for this length (80% of critical speed)?

a) Travel = $36 \frac{in}{cycle} \times 12 \frac{cycles}{hour} \times 24 \frac{hours}{day} \times 365 \frac{days}{year} \times 10\ years$

$= 37\,843\,200\ in = 3.8 \times 10^7\ in$ (1 051 200 cycles)

[Fig. 17-8]:

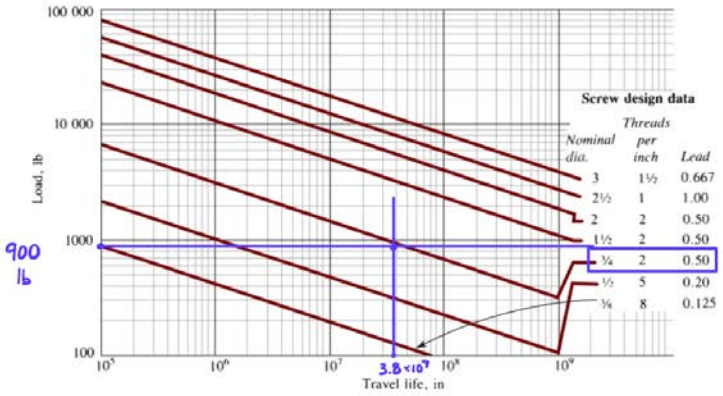


FIGURE 17–8 Ball screw performance

Select: 3/4 in ø, 2 threads/inch, 0.50 in Lead

b) [Eq. 17-13]

$$T = \frac{FL}{2\pi e} = \frac{900\ lb\ (0.5\ in)}{2\pi\ (0.90)} = 79.58\ lb\text{-}in$$

(assume $e = 0.90$)

$T = 79.6\ lb\text{-}in$ TO DRIVE LOAD

c) $2 \frac{in}{sec} \times \frac{1\ rev}{0.5\ in} \times \frac{60\ s}{min} = 240\ rpm$ required

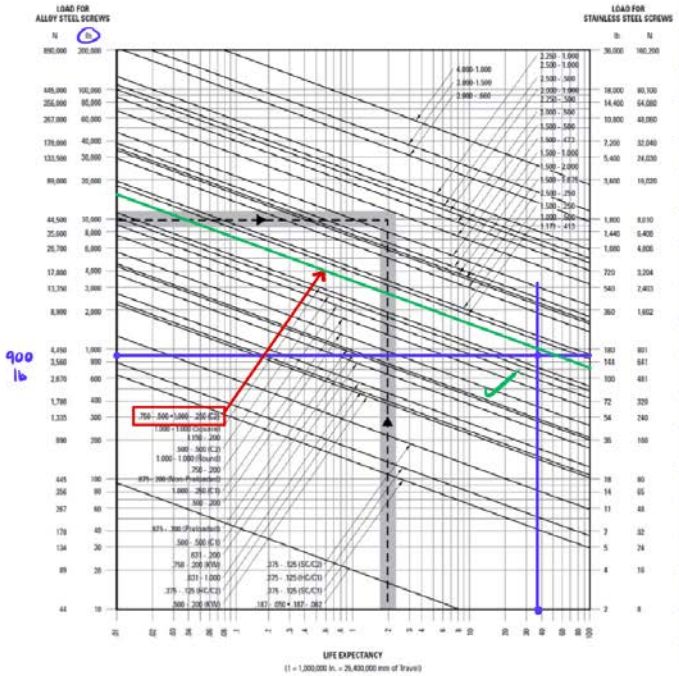
$$P = \frac{T \cdot n}{63025} = \frac{79.58\ lb\text{-}in\ (240\ rpm)}{63025} = 0.303\ hp$$

$P = 0.3\ hp$

d) Thompson Guide

Engineering Guidelines for Inch Series Ball Screws

Life Expectancy for Precision Ball Screw Assemblies



OK for life expectancy ✓

- Thompson agrees with Mott selection

$L = 37.8 \times 10^6$ in

e) Both ends fixed

S.F. 2.5

MAXIMUM LENGTH?

→ effective load = 900 × S.F.

= 2250 lb

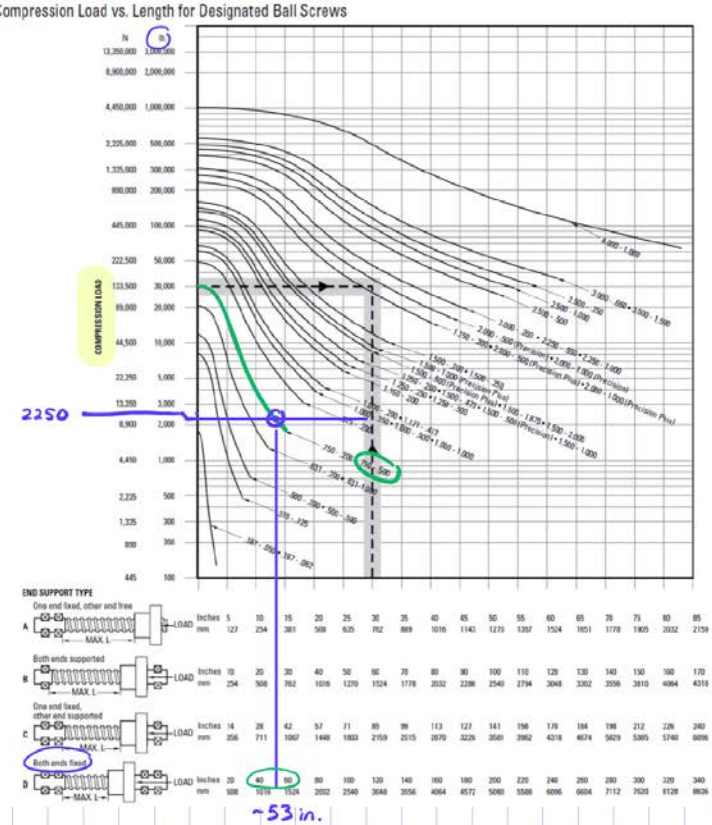
$L_{max} \approx 53$ in

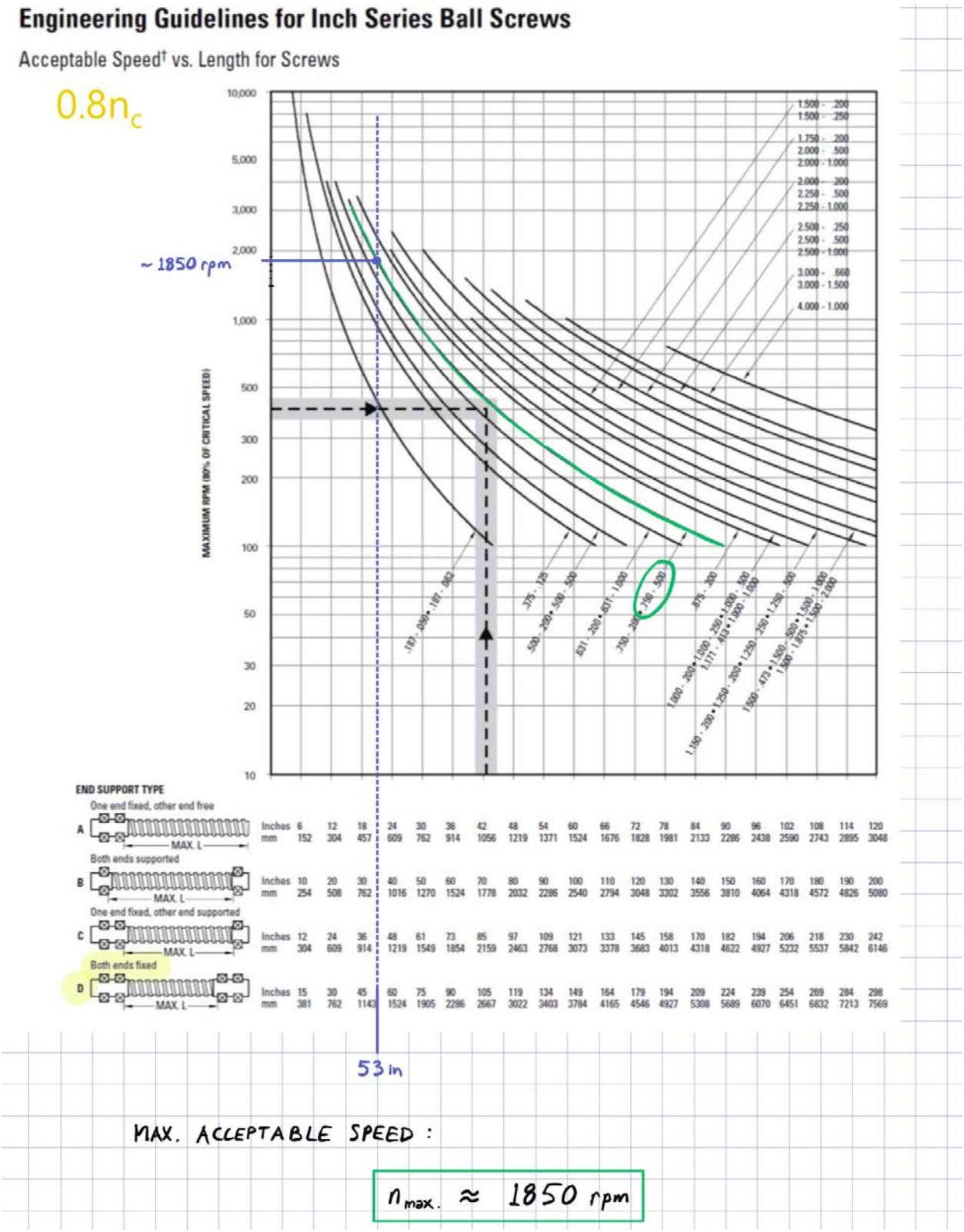
MAXIMUM SPEED?



Engineering Guidelines for Inch Series Ball Screws

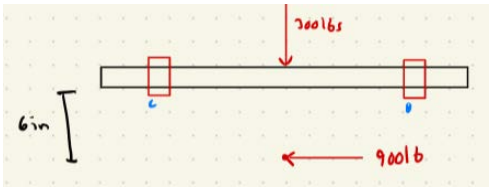
Compression Load vs. Length for Designated Ball Screws





f) The platform is supporting the weight of 300 lbs at the midpoint of all four linear bearings. Hence each bearing will support 75lbs of weight.

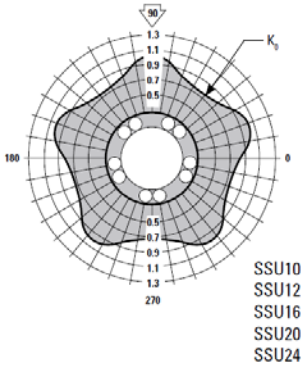
The axial working force of 900lbs occurs 6 inches below the CG of the platform. The moment caused by this force M_y = 900(6) = 5400 in-lbs. This moment is resisted by two pairs of linear bearings on the platform (fore and aft bearings) that are separated by 8inches.



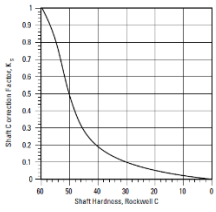
Hence the force on each pair of bearings is 5400/8 = 675 lbs. The force on the forward bearings act downward while the force on the rear bearings act upwards.

Total force on each forward bearing F_b = (675/2)+75 = 412.5lbs.

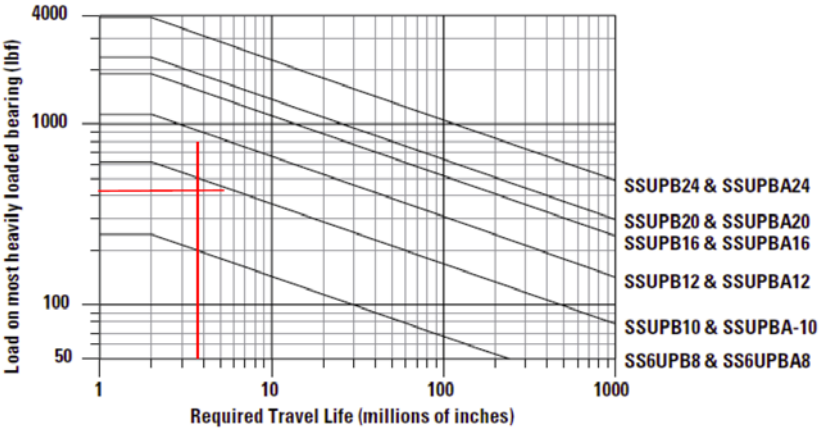
This force is vertical so the load distribution factor, K_o = 1.0. Let's assume the rails are set to a hardness of at least HRC = 60 so that the shaft correction factor, K_s = 1.0



$$W_R = \frac{P}{K_0 \bullet K_S \bullet K_L}$$

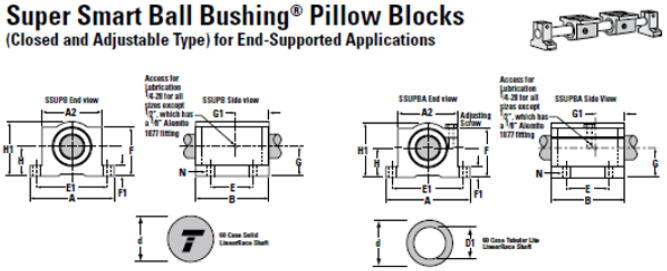


Use Figure from Thomson Linear Bearings at 37 (10^6)inches and 412 lbs falls just below the line for SSUPB10. The rating for this linear bearing is approximately 500lbs for 37(10^6)inches of travel giving a safety factor of 500/412 = 1.21 for fatigue. Hence the rail size must be 5/8" diameter for this application.



THOMSON
Linear Motion. Optimized.

Super Smart Ball Bushing® Pillow Blocks
(Closed and Adjustable Type) for End-Supported Applications



Super Smart Ball Bushing Pillow Blocks (Closed & Adjustable Types, seal at both ends) and LinearRace® (Dim. in inches)

Part Number[1]			Nominal Diameter	H ±.003	H1	60 Case LinearRace Diameter d	60 Case LinearRace Minimum Depth of Hardness	60 Case Solid LinearRace Mass lb/in	60 Case Tubular Lite LinearRace Mass lb/in	60 Case Tubular Lite LinearRace ID D1
Super Smart Ball Bushing Pillow Block		60 Case LinearRace								
Fixed	Adjustable									
SSBUPB8	SSBUPBA8	1/2 L	.500	.687	1.25	.4995/.4990	.04	.06	–	–
SSUPB10	SSUPBA10	5/8 L	.625	.875	1.63	.6245/.6240	.04	.09	–	–
SSUPB12	SSUPBA12	3/4 L	.750	.937	1.75	.7495/.7490	.06	.13	.08	.66/.41
SSUPB16	SSUPBA16	1 L	1.000	1.187	2.19	.9995/.9990	.08	.22	.16	.62/.56
SSUPB20	SSUPBA20	1 1/4 L	1.250	1.500	2.81	1.2495/1.2490	.08	.35	–	–
SSUPB24	SSUPBA24	1 1/2 L	1.500	1.750	3.25	1.4994/1.4989	.08	.50	.33	.93/.84

Part Number[1]		A	A2	B	E ±.010	E1 ±.010	F	F1	G	G1	N		Pillow Block Mass lb	Dynamic[2] Load Capacity lb$_f$
Super Smart Ball Bushing Pillow Block											Hole	Bolt		
Fixed	Adjustable													
SSBUPB8	SSBUPBA8	2.00	1.38	1.69	1.000	1.688	1.13	.25	.97	.44	.16	#6	.23	265
SSUPB10	SSUPBA10	2.50	1.75	1.94	1.125	2.125	1.44	.28	1.20	.69	.19	#8	.51	625
SSUPB12	SSUPBA12	2.75	1.88	2.06	1.250	2.375	1.56	.31	.94	.78	.19	#8	.62	1130
SSUPB16	SSUPBA16	3.25	2.38	2.81	1.750	2.875	1.94	.38	1.19	.91	.22	#10	1.24	1900
SSUPB20	SSUPBA20	4.00	3.00	3.63	2.000	3.500	2.50	.44	1.50	1.37	.22	#10	2.57	2950
SSUPB24	SSUPBA24	4.75	3.50	4.00	2.500	4.125	2.88	.50	1.75	1.13	.28	.25	3.94	3880

One can also estimate the frictional drag for the bearings. If one assumes grease lubrication to help prevent corrosion, we find that for 5/8” diameter and a rating of 412/500 = 0.8 which is close to 75% of rating has a coefficient of friction of 0.0024.

Friction force forward bearings = 412(0.0024) = 0.998lbs = 1.0lb. The entire friction drag will be less than 4 lbs which is negligible.

Ball Bushing Bearing coefficients of rolling friction (f_r)

Bearing I.D.	Number of Ball Circuits	Condition of Lubrication	Load in % of Rolling Load Rating (for 2,000,000 inches of travel)				
			125%	100%	75%	50%	25%
1/4, 3/8, 1/2, 5/8	3 & 4	No Lube	.0011	.0011	.0012	.0016	.0025
		Grease Lube	.0019	.0021	.0024	.0029	.0044
		Oil Lube	.0022	.0023	.0027	.0032	.0045
3/4, 1	5	No Lube	.0011	.0011	.0012	.0015	.0022
		Grease Lube	.0018	.0019	.0021	.0024	.0033
		Oil Lube	.0020	.0021	.0023	.0027	.0036
1 1/4 thru 1	6	No Lube	.0011	.0011	.0012	.0014	.0019
		Grease Lube	.0016	.0016	.0017	.0018	.0022
		Oil Lube	.0018	.0018	.0019	.0021	.0027
5/8 thru 1 1/2	10	No Lube	.0011	.0011	.0012	.0013	.0018
		Grease Lube	.0014	.0014	.0015	.0016	.0019
		Oil Lube	.0016	.0016	.0017	.0019	.0025

Problem 6 (Question 10-6)

A helical compression spring is to be made of oil-tempered wire of 4-mm diameter with a spring index of $C = 10$. The spring is to operate inside a hole, so buckling is not a problem and the ends can be left plain. The free length of the spring should be 80 mm. A force of 50 N should deflect the spring 15 mm.

(*a*) Determine the spring rate.

(*b*) Determine the minimum hole diameter for the spring to operate within.

(c) Determine the total number of coils needed.

(*d*) Determine the solid length.

(*e*) Determine a static factor of safety based on the yielding of the spring if it is compressed to its solid length.

Solution 10-6

Given: Oil-tempered wire, $d = 4$ mm, $C = 10$, plain ends, $L_0 = 80$ mm, and at $F = 50$ N, $y = 15$ mm.

(**a**) $k = F/y = 50/15 = 3.333$ N/mm *Ans.*

(**b**) $D = Cd = 10(4) = 40$ mm

OD $= D + d = 40 + 4 = 44$ mm *Ans.*

(**c**) From Table 10-5, $G = 77.2$ GPa

Eq. (10-9): $$N_a = \frac{d^4 G}{8kD^3} = \frac{4^4(77.2)10^3}{8(3.333)40^3} = 11.6 \text{ coils}$$

Table 10-1: $N_t = N_a = 11.6$ coils *Ans.*

(**d**) Table 10-1: $L_s = d(N_t + 1) = 4(11.6 + 1) = 50.4$ mm *Ans.*

(**e**) Table 10-4: $m = 0.187$, $A = 1855$ MPa·mm^m

Eq. (10-14): $$S_{ut} = \frac{A}{d^m} = \frac{1855}{4^{0.187}} = 1431 \text{ MPa}$$

Table 10-6: $S_{sy} = 0.50\, S_{ut} = 0.50(1431) = 715.5$ MPa

$$y_s = L_0 - L_s = 80 - 50.4 = 29.6 \text{ mm}$$

$$F_s = k\, y_s = 3.333(29.6) = 98.66 \text{ N}$$

Eq. (10-5): $$K_B = \frac{4C+2}{4C-3} = \frac{4(10)+2}{4(10)-3} = 1.135$$

Eq. (10-7): $$\tau_s = K_B \frac{8F_s D}{\pi d^3} = 1.135 \frac{8(98.66)40}{\pi(4^3)} = 178.2 \text{ MPa}$$

$$n_s = \frac{S_{sy}}{\tau_s} = \frac{715.5}{178.2} = 4.02 \qquad \textit{Ans.}$$
